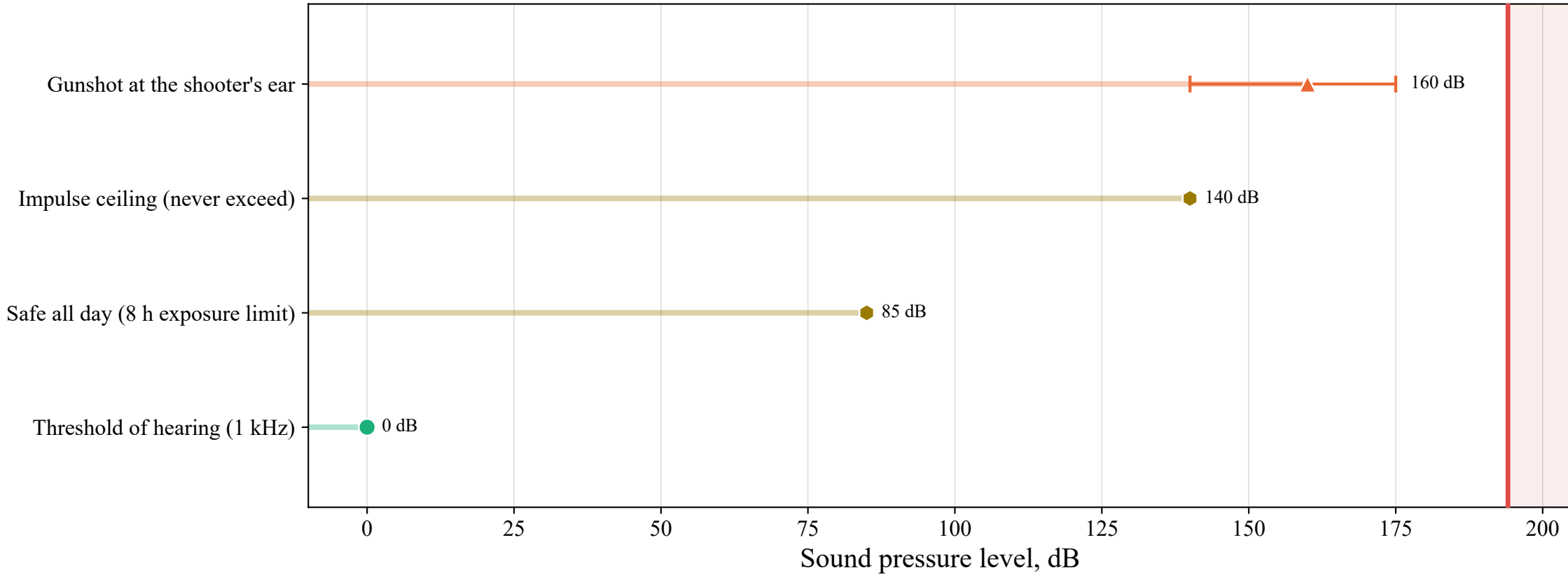
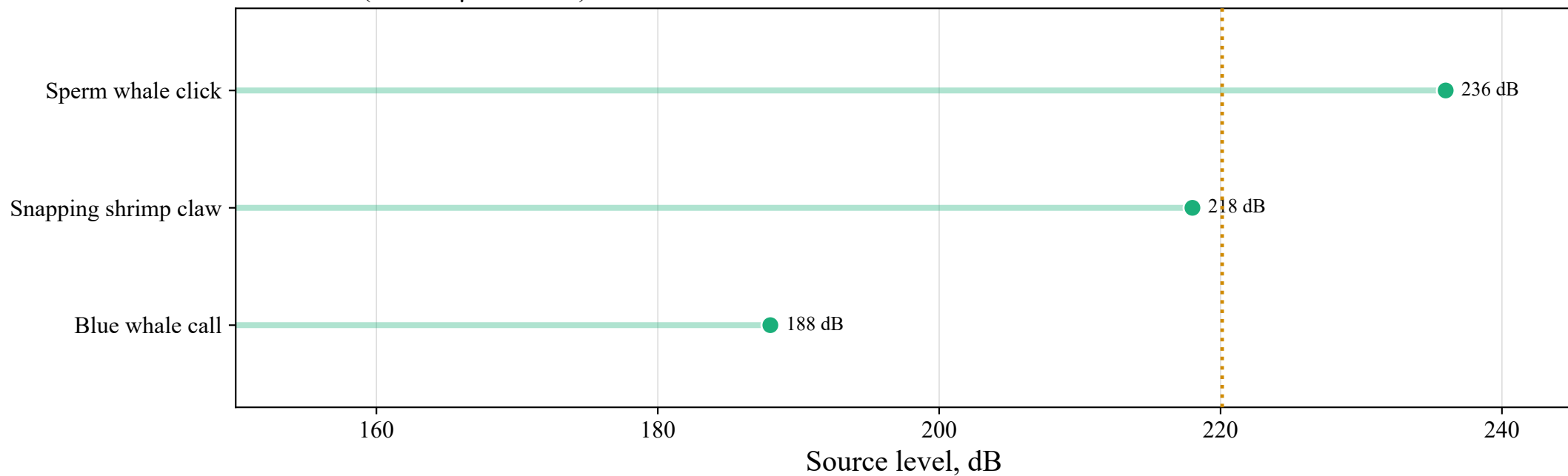


How loud can sound be?

a. In air (dB re 20 μ Pa)



b. In water (dB re 1 μ Pa at 1 m)



- Animals and hearing (teal circle)
- Exposure limits (olive diamond)
- Human-made (orange triangle)
- Reported range (grey line with error bars)
- 194 dB in air: the trough reaches vacuum (amplitude = 1 atm) (red line)
- 220 dB: cavitation near the surface (deeper water allows more) (orange dotted line)

Air and water levels use different reference pressures and cannot be compared directly.